Nahian Tahmin, Ph.D.

Computational Biologist | Bioinformatics and Machine Learning for Genomics

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PROFESSIONAL SUMMARY

Computational biology and machine learning researcher specializing in multimodal genomic prediction and pharmacogenomics. Builds interpretable deep learning and statistical genetics models that integrate genotype, local ancestry, and social drivers of health (SDOH), with a focus on admixed populations, precision medicine, and health equity. First-author publication in Scientific Reports and a preprint under review at PLOS Computational Biology. Software engineering training supports reproducible, scalable bioinformatics pipelines in Python, R, and PyTorch.

EDUCATION

University of Tennessee, Knoxville Aug 2022 - Aug 2026

Ph.D., Data Science and Engineering. GPA 3.95/4.00. Advisor: Dr. Anahita Khojandi.

Monash University Malaysia Oct 2017 - Oct 2021

B.Eng., Software Engineering.

PUBLICATIONS

Tahmin N, Chinthala L, Marsico F, Buonaiuto S, Mohammed A, Carlisle A, Gautam Y, Colonna V, Mersha T, Davis RL, Khojandi A. Machine learning models incorporating genotype and ancestry improve severe asthma risk prediction. *Scientific Reports* 15, 40243 (2025). doi.org/10.1038/s41598-025-24080-x

Tahmin N, Chinthala LK, Mersha TB, Davis RL, Khojandi A. X-Admix: An Interpretable Multimodal Cross-Attention Framework for Integrating Genotype, Local Ancestry, and Social Drivers of Health in Admixed African American Populations. *medRxiv* preprint (2026); under review at PLOS Computational Biology. doi.org/10.64898/2026.09.03.26362179

Pokhrel D, **Tahmin N**, Carlisle A, Mersha T, Davis RL, Khojandi A. Assessing the Role of Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis. *IEEE SIEDS* (2025). doi.org/10.1109/SIEDS65500.2025.11021139

Predicting Precision Inhaled Corticosteroids Response in Asthma Patients. Presented at the ASHG Annual Meeting 2023, Abstract 2023-A-1763-ASHG.

Integrating genomics, local ancestry, and social drivers of health (SDOH) on asthma medication response prediction. Presented at ASHG Annual Meeting 2023 and INFORMS Annual Meeting 2024 and 2025.

PROJECTS

Multimodal ML/DL/LLM Pipelines for Genotype, Local Ancestry, and SDOH in Asthma Prediction Aug 2022 - Jul 2026

University of Tennessee, Knoxville

- **Multimodal deep learning (X-Admix).** Designed and built a framework that fuses genotype, local ancestry, and SDOH through pairwise cross-attention and a learned softmax-gated fusion layer, capturing conditional cross-modal interactions that additive models miss. Reached AUC 0.771 on a low-signal asthma prediction task in an admixed African American cohort, where classical baselines stayed near chance (AUC 0.566 or lower). The performance trend held on a second, independent All of Us cohort roughly 3x larger.
- **Explainable AI for biological discovery.** Layered feature ablation, gradient-based attribution, and SHAP with a leave-one-component-out protocol. Designed a two-tier consensus strategy that ranks candidate cross-modal interactions by cross-fold reproducibility, separating a robust, ablation-stable core from a larger discovery pool. Showed that a component can be redundant for accuracy yet still determine which interactions the model surfaces.
- **Pharmacogenomic treatment response prediction.** Integrated genotype and local ancestry in a pediatric asthma cohort (N = 248, about 5 million features) to predict inhaled corticosteroid (ICS) response. Combined Lasso, Recursive Feature Elimination (RFE), and Elastic Net feature selection with expert priors; the stacked Lasso and Random Forest model beat unimodal baselines by 5 AUC points, and ablations showed the modalities carry non-overlapping signal.
- **Genomic language models (completed as part of doctoral dissertation, 2026).** Compared four serialization schemes for encoding genomic variant data for pretrained large language models (LLMs) under a fixed token budget in an admixed All of Us cohort. Compressed encodings outperformed verbose prose descriptions (AUC 0.71 vs. 0.63). Designed an embedding-injection scheme that extends to unseen variants without adding parameters.

Advisor, Vertically Integrated Program (VIP) Team, Criminal Sentences Project, *Human-Augmented Analytics Group*

- Advised a cross-functional student team building Named Entity Recognition (NER) models that extract sentencing components from federal court records. Supervised evaluation criteria and error analysis, and mentored the team on problem framing, data preparation, and LLM-based labeling automation.

Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis

- Mentored a junior Ph.D. student to build interpretable models integrating demographic, socioeconomic, and clinical features, using targeted ablations to identify subgroup-specific SDOH drivers. Oversaw end-to-end study design; published at IEEE SIEDS 2025.

SKILLS

Genomics and Bioinformatics: PLINK, RFMix, SHAPEIT, All of Us Researcher Workbench, genotype data, local ancestry inference, admixed cohorts, pharmacogenomics, whole-genome sequencing (WGS) data, exome-matched variant filtering

Machine Learning: PyTorch, Transformers, cross-attention, Hugging Face, Large Language Models (LLM), PEFT/LoRA, scikit-learn, TensorFlow, Captum, SHAP

Statistics and Modeling: Lasso, Elastic Net, Random Forest, ensemble stacking, feature selection, ablation studies, cross-validation, hyperparameter tuning

Programming and Data: Python, R, SQL, PySpark, Pandas, NumPy, Java, C, C++, MATLAB, Git/GitHub, Google Cloud, Weights and Biases

PROFESSIONAL EXPERIENCE

University of Tennessee, Knoxville, *Graduate Research Assistant* Aug 2022 - Jul 2026

Designed scalable, reliable, and interpretable AI, deep learning, and LLM research workflows; data engineering and fusion for multimodal genomic datasets; worked in a cross-disciplinary team of clinicians, geneticists, and engineers.

ITQAN Analytics and Software Limited, *Developer Intern* Mar 2022 - Jul 2022

Developed Python data engineering pipelines and Power BI dashboards for data visualization.

Monstar Lab Bangladesh, *Frontend Intern* Dec 2019 - Feb 2020

Developed chatbot interfaces using AngularJS and Node.js with senior mentors.